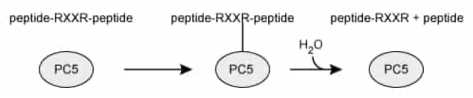
Proprotein convertases (PCs) are serine proteases that cleave inactive proproteins at the surface of cells, converting them into active proteins. The proprotein convertase known as PC5 is critical for development, and knocking out PC5 during embryonic development is lethal in mice. After PC5 arrives at the cell surface, it first processes itself by autocleavage (ie, cleaving itself) before acting on any other proteins.

Many PCs cleave proteins at sites with the amino acid sequence RXXR, where X is any amino acid. In PC5, the side chain of a serine residue in the active site becomes deprotonated and acts as a nucleophile to form a covalent bond with an arginine residue, eventually hydrolyzing the protein at the C-terminal side of the bound arginine residue, as shown in Scheme A.

Scheme A



To characterize PC5 activity, a growth factor protein called Lefty was expressed in Chinese Hamster Ovary (CHO) cells. The inactive form of Lefty (Prolefty) has two PC5 cleavage sites, as illustrated in Figure 1. Depending on which site or sites PC5 cleaves, Prolefty may be converted into one of two Lefty isoforms: the long form (L) or the short form (S). Relative amounts of each form of Lefty, either in the absence or presence of PC5 coexpression, were measured in CHO cells.

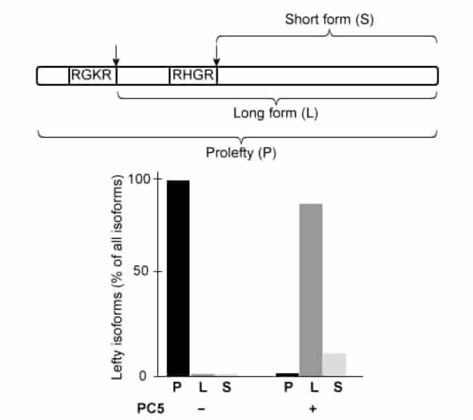
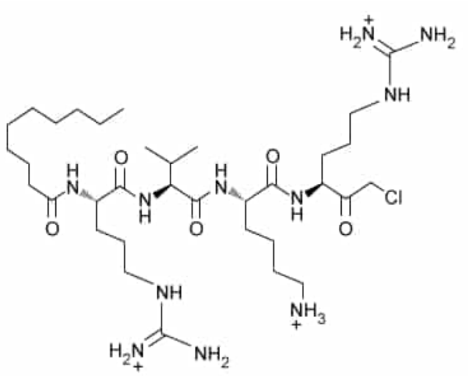


Figure 1 Schematic of Lefty (top) and graphical results of relative levels of each Lefty isoform detected by Western blot in the absence (−) or presence (+) of PC5 (bottom). Arrows in the top panel indicate cleavage sites.

Some chloromethylketone (CMK) derivatives are known to inhibit another PC called furin.



微信Mai_ange

Figure 2 Structure of a CMK molecule used to inhibit furin.

In a separate experiment, PC5 activity against Lefty was assessed in a mouse cell line, either in the absence or presence of a specific CMK that inhibits furin (Figure 3).

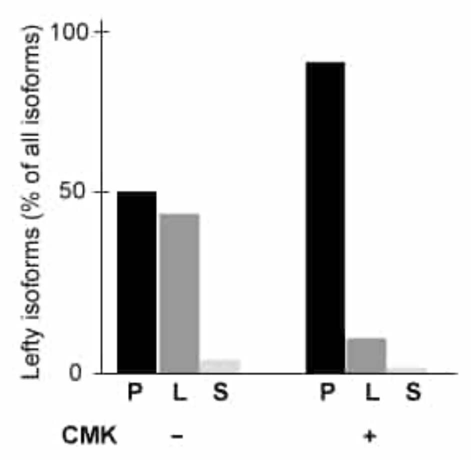


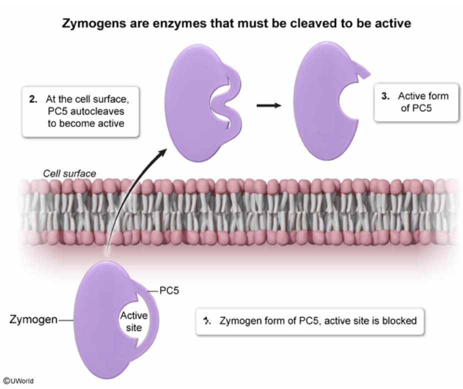
Figure 3 Graphical results of the relative amounts of each Lefty isoform in mouse cells expressing PC5 in the absence (−) or presence (+) of the furin-inhibiting CMK.

Prior to reaching the cell surface, PC5 is:

- A. an apoenzyme [17%]
- B. a coenzyme [8%]
- C. a holoenzyme [8%]
- ✔ D. a zymogen [64%]

Correct
64% answered correctly

Explanation:



Some enzymes are initially synthesized in an **inactive form** to prevent them from acting on potential substrates in the wrong cellular compartment. These enzymes, known as **zymogens**, are **activated when a portion** of their polypeptide chain is **cleaved**. Often the cleaved portion was previously blocking access to the enzyme active site. Some zymogens are activated when other enzymes catalyze the cleavage reaction, and other zymogens autocatalyze their own cleavage in response to environmental cues.

The passage states that PC5 processes itself by autocleavage when it arrives at the cell surface and that it does not act on other proteins until after autocleavage occurs. Therefore, prior to reaching the cell surface, **PC5 is a zymogen**.

(Choices A and C) Apoenzymes and holoenzymes are two forms of the same enzyme. In the apo form, the enzyme is not bound to a cofactor and is inactive, whereas the holo form is bound to its cofactor (eg, a metal ion) and is active. There is no indication in the passage that PC5 requires a cofactor; therefore, there is insufficient information about whether it can be characterized as an apo- or holoenzyme.

(Choice B) A coenzyme is a nonprotein molecule that is required for an enzyme to function. PC5 is a protein, so it cannot be a coenzyme.

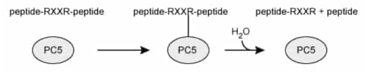
Educational objective:
Zymogens are enzymes that are rendered inactive by a portion of their own amino acid sequence. Zymogens are activated by cleavage of this sequence, either by their own action or by the action of another protein.

Time spent:0 sec	**QID**:403358
Subject:	**System:**
Biochemistry	Enzymes

Proprotein convertases (PCs) are serine proteases that cleave inactive proproteins at the surface of cells, converting them into active proteins. The proprotein convertase known as PC5 is critical for development, and knocking out PC5 during embryonic development is lethal in mice. After PC5 arrives at the cell surface, it first processes itself by autocleavage (ie, cleaving itself) before acting on any other proteins.

Many PCs cleave proteins at sites with the amino acid sequence RXXR, where X is any amino acid. In PC5, the side chain of a serine residue in the active site becomes deprotonated and acts as a nucleophile to form a covalent bond with an arginine residue, eventually hydrolyzing the protein at the C-terminal side of the bound arginine residue, as shown in Scheme A.

Scheme A



To characterize PC5 activity, a growth factor protein called Lefty was expressed in Chinese Hamster Ovary (CHO) cells. The inactive form of Lefty (Prolefty) has two PC5 cleavage sites, as illustrated in Figure 1. Depending on which site or sites PC5 cleaves, Prolefty may be converted into one of two Lefty isoforms: the long form (L) or the short form (S). Relative amounts of each form of Lefty, either in the absence or presence of PC5 coexpression, were measured in CHO cells.

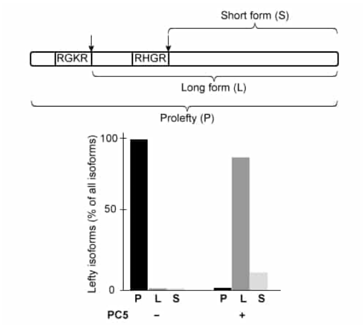


Figure 1 Schematic of Lefty (top) and graphical results of relative levels of each Lefty isoform detected by Western blot in the absence (−) or presence (+) of PC5 (bottom). Arrows in the top panel indicate cleavage sites.

Some chloromethylketone (CMK) derivatives are known to inhibit another PC called furin.

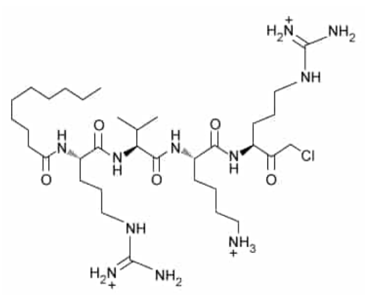


Figure 2 Structure of a CMK molecule used to inhibit furin.

In a separate experiment, PC5 activity against Lefty was assessed in a mouse cell line, either in the absence or presence of a specific CMK that inhibits furin (Figure 3).

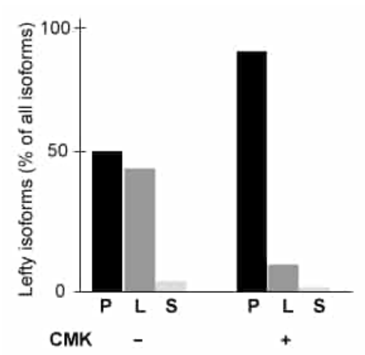


Figure 3 Graphical results of the relative amounts of each Lefty isoform in mouse cells expressing PC5 in the absence (−) or presence (+) of the furin-inhibiting CMK.

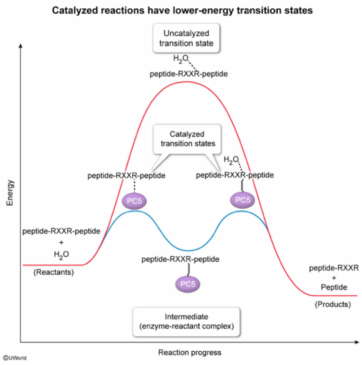
Compared to uncatalyzed protein hydrolysis, formation of a covalent bond between the active-site serine of PC5 and an arginine residue in a substrate protein most likely results in which of the following?

- ◯ A. A more stable set of reaction products. [12%]
- ◯ B. A more negative free energy change of the overall hydrolysis reaction. [12%]

- [x] C. A lower-energy transition state when water reacts with the substrate protein. [68%]
- [ ] D. An increased number of hydrolyzed proteins once the reaction reaches equilibrium. [6%]

Correct
69% answered correctly

Explanation:



Enzymes catalyze reactions when the enzyme active site interacts with the reacting molecule(s) (ie, substrates). These interactions often involve temporary formation of covalent bonds between a substrate molecule and one or more amino acid side chains in the active site, which **facilitates later steps** in the reaction mechanism. Regardless of the details of the mechanism, all enzymes **increase the rate of a reaction** by providing a pathway with lower-energy transition states than would be possible in the absence of the enzyme.

The information in the passage text and in Scheme A indicate that the side chain of a serine residue forms a covalent bond with arginine in a substrate protein, and water subsequently hydrolyzes the substrate protein. Proteins can also be hydrolyzed in the absence of enzymes, but this process is very slow. Serine proteases such as PC5 increase hydrolysis rates by forming covalent bonds to the peptide to be hydrolyzed, thereby decreasing the energy of the transition state for the hydrolysis reaction.

(Choice A) The stability of a reaction's products depends only on the identity of those products, *not* on the mechanism by which they form.

(Choice B) The free energy change of a reaction is a function of state and depends only on the identities of the products and reactants. The presence or absence of an enzyme does not alter the free energy change of the reaction.

(Choice D) Enzymes do not change the equilibrium position of a reaction; they only alter how quickly equilibrium is achieved. Thus, a catalyzed reaction may *temporarily* have more products than its uncatalyzed counterpart before equilibrium is reached, but once both have reached equilibrium, they will have the same number of products.

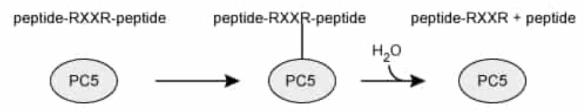
Educational objective:
Enzymes increase reaction rates by providing a reaction pathway with lower-energy transition states than the uncatalyzed reaction. Enzymes do *not* alter product stability, Gibbs free energy, or reaction equilibrium.

Time spent:0 sec	**QID:**403359
Subject:	**System:**
Biochemistry	Enzymes

Proprotein convertases (PCs) are serine proteases that cleave inactive proproteins at the surface of cells, converting them into active proteins. The proprotein convertase known as PC5 is critical for development, and knocking out PC5 during embryonic development is lethal in mice. After PC5 arrives at the cell surface, it first processes itself by autocleavage (ie, cleaving itself) before acting on any other proteins.

Many PCs cleave proteins at sites with the amino acid sequence RXXR, where X is any amino acid. In PC5, the side chain of a serine residue in the active site becomes deprotonated and acts as a nucleophile to form a covalent bond with an arginine residue, eventually hydrolyzing the protein at the C-terminal side of the bound arginine residue, as shown in Scheme A.

Scheme A



To characterize PC5 activity, a growth factor protein called Lefty was expressed in Chinese Hamster Ovary (CHO) cells. The inactive form of Lefty (Prolefty) has two PC5 cleavage sites, as illustrated in Figure 1. Depending on which site or sites PC5 cleaves, Prolefty may be converted into one of two Lefty isoforms: the long form (L) or the short form (S). Relative amounts of each form of Lefty, either in the absence or presence of PC5 coexpression, were measured in CHO cells.

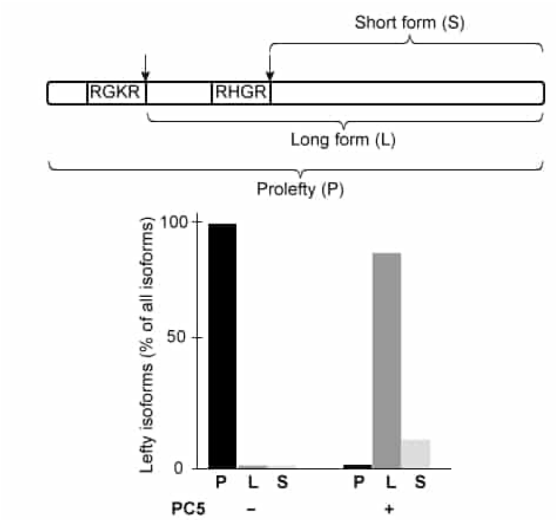


Figure 1 Schematic of Lefty (top) and graphical results of relative levels of each Lefty isoform detected by Western blot in the absence (−) or presence (+) of PC5 (bottom). Arrows in the top panel indicate cleavage sites.

Some chloromethylketone (CMK) derivatives are known to inhibit another PC called furin.

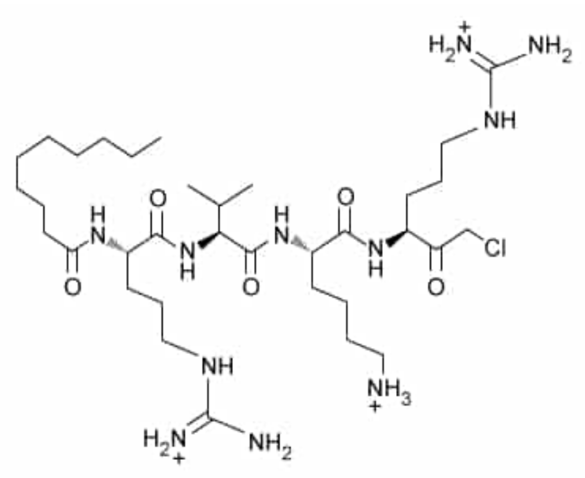


Figure 2 Structure of a CMK molecule used to inhibit furin.

In a separate experiment, PC5 activity against Lefty was assessed in a mouse cell line, either in the absence or presence of a specific CMK that inhibits furin (Figure 3).

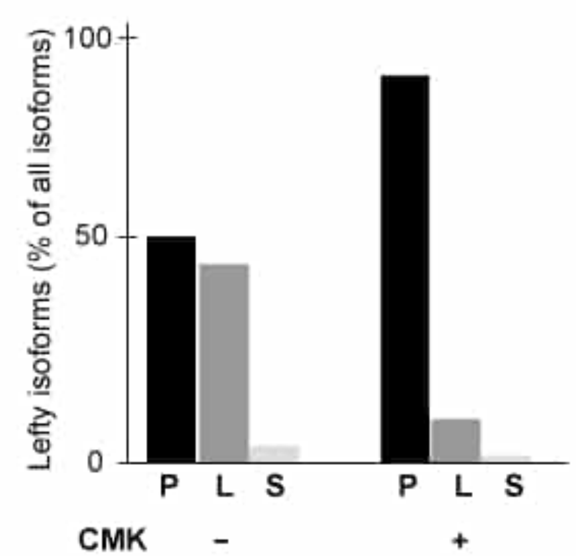
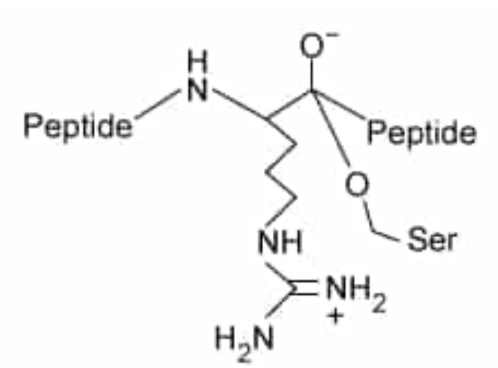


Figure 3 Graphical results of the relative amounts of each Lefty isoform in mouse cells expressing PC5 in the absence (−) or presence (+) of the furin-inhibiting CMK.

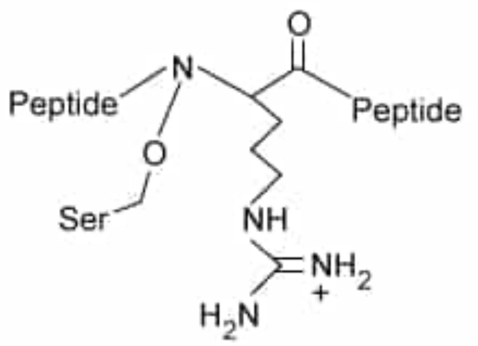
Which of the following represents an intermediate formed in the PC5-catalyzed hydrolysis of a peptide?

✔ A.



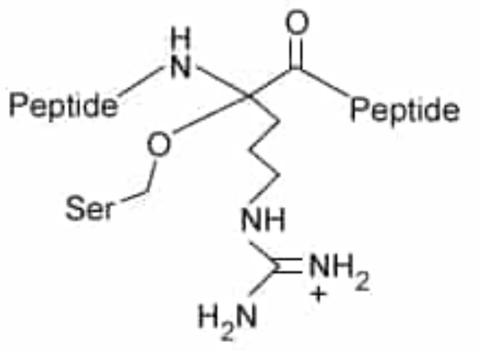
[51%]

B.



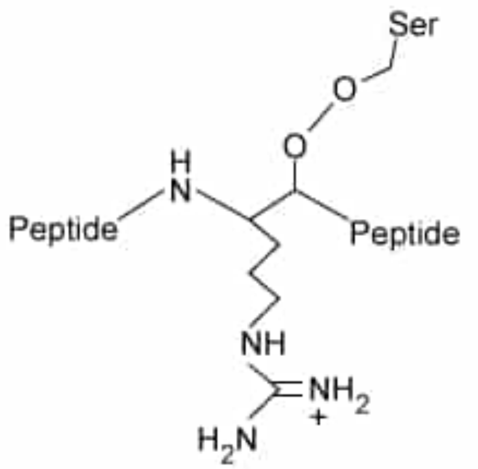
[11%]

C.



[29%]

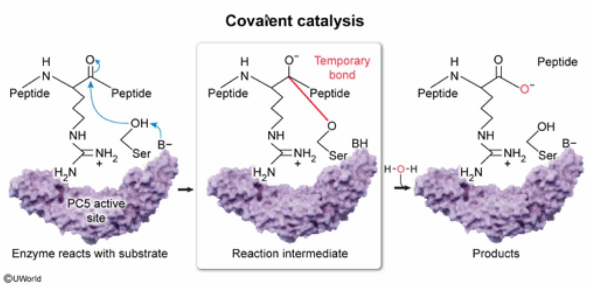
D.



[7%]

Correct
51% answered correctly

Explanation:



Covalent catalysis occurs when an enzyme **temporarily forms one or more covalent bonds with its substrate** as part of the mechanism that eventually converts the substrate to product. This commonly occurs when a nucleophile in the enzyme active site attacks an electrophile in the substrate molecule. In serine proteases, an active site serine side chain becomes deprotonated, allowing it to act as the nucleophile.

An amino acid residue is an amino acid from which H_2O has been removed upon incorporation into a peptide. The passage states that a serine residue acts as a nucleophile to attack an arginine residue. Therefore, serine must attack an electrophilic portion of the arginine residue (ie, an atom with a partial positive charge). The carbonyl carbon in the backbone of any amino acid residue is an electrophile because the oxygen atom to which it is bonded pulls electron density away.

Based on this information, the side chain of the serine residue can act as a nucleophile to attack the carbonyl carbon in the backbone of the arginine residue, forming the intermediate shown in Choice A. Additional steps, including nucleophilic attack by water, then cleave the substrate peptide.

(Choice B) The nitrogen atom in the amino acid backbone is a poor electrophile because it is more electronegative than the atoms to which it is bonded, and it pulls electron density toward itself. Serine is unlikely to attack this atom.

(Choice C) This diagram shows the result of serine attacking the alpha carbon of arginine (ie, the carbon to which the side chain is attached). This mechanism would require removal of the hydrogen atom that is normally bonded to the alpha carbon so as not to have 5 bonds to carbon. Such a mechanism is unlikely.

(Choice D) This diagram depicts the intermediate that would form if serine attacked the carbonyl *oxygen* of arginine rather than the carbonyl carbon. The carbonyl oxygen has a partial negative charge and is a poor electrophile.

Educational objective:
Covalent catalysis occurs when an amino acid in the active site of an enzyme forms a temporary covalent bond with the substrate as part of the reaction mechanism. This generally occurs when an active site residue acts as a nucleophile to attack an electrophile in the substrate.

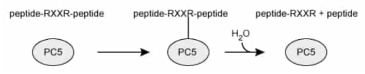
Time spent:0 sec
Subject:
Biochemistry

QID:403360
System:
Enzymes

Proprotein convertases (PCs) are serine proteases that cleave inactive proproteins at the surface of cells, converting them into active proteins. The proprotein convertase known as PC5 is critical for development, and knocking out PC5 during embryonic development is lethal in mice. After PC5 arrives at the cell surface, it first processes itself by autocleavage (ie, cleaving itself) before acting on any other proteins.

Many PCs cleave proteins at sites with the amino acid sequence RXXR, where X is any amino acid. In PC5, the side chain of a serine residue in the active site becomes deprotonated and acts as a nucleophile to form a covalent bond with an arginine residue, eventually hydrolyzing the protein at the C-terminal side of the bound arginine residue, as shown in Scheme A.

Scheme A



To characterize PC5 activity, a growth factor protein called Lefty was expressed in Chinese Hamster Ovary (CHO) cells. The inactive form of Lefty (Prolefty) has two PC5 cleavage sites, as illustrated in Figure 1. Depending on which site or sites PC5 cleaves, Prolefty may be converted into one of two Lefty isoforms: the long form (L) or the short form (S). Relative amounts of each form of Lefty, either in the absence or presence of PC5 coexpression, were measured in CHO cells.

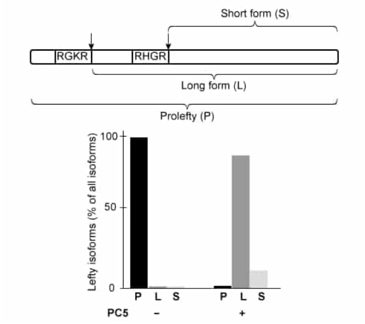


Figure 1 Schematic of Lefty (top) and graphical results of relative levels of each Lefty isoform detected by Western blot in the absence (−) or presence (+) of PC5 (bottom). Arrows in the top panel indicate cleavage sites.

Some chloromethylketone (CMK) derivatives are known to inhibit another PC called furin.

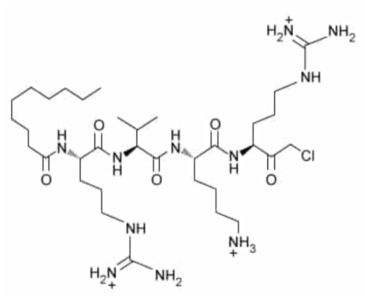


Figure 2 Structure of a CMK molecule used to inhibit furin.

In a separate experiment, PC5 activity against Lefty was assessed in a mouse cell line, either in the absence or presence of a specific CMK that inhibits furin (Figure 3).

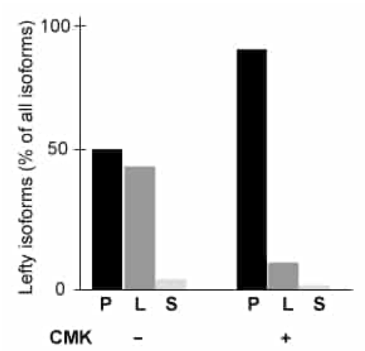


Figure 3 Graphical results of the relative amounts of each Lefty isoform in mouse cells expressing PC5 in the absence (−) or presence (+) of the furin-inhibiting CMK.

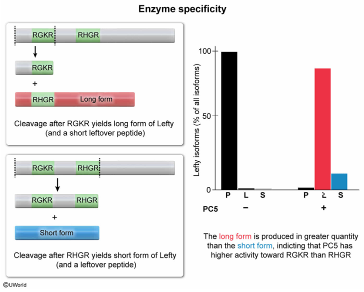
Which of the following statements regarding the specificity of PC5 is best supported by the data in Figure 1?

- A. PC5 has activity toward the sequence RGKR but not toward the sequence RHGR. [1%]
- ✔ B. PC5 has a higher activity toward the sequence RGKR than toward the sequence RHGR. [71%]
- C. PC5 has a higher activity toward the sequence RHGR than toward the sequence RGKR. [22%]

D. PC5 has approximately equal activity toward the sequences RGKR and RHGR. [3%]

Correct
71% answered correctly

Explanation:



Enzymes may have **different levels of specificity**. Some enzymes interact only with one substrate or a few similar substrates. Other enzymes act on multiple substrates with broadly similar characteristics and may have higher activity toward some substrates than others. This specificity can be measured empirically.

Figure 1 shows that in the presence of PC5, the protein Prolefty is almost entirely converted into either the long or the short isoform of active Lefty. The top portion of the figure shows that to generate the long form, PC5 must cleave Prolefty after the amino acid sequence RGKR, whereas the short form is generated by cleaving after the sequence RHGR. The graph in Figure 1 shows that the long form is generated in substantially larger amounts than the short form, indicating that PC5 has a higher activity toward the RGKR sequence than the RHGR sequence.

(Choice A) If PC5 had no activity toward RHGR, the short form of Lefty would not be produced at all. Figure 1 shows that the short form is produced to a small extent.

(Choice C) If PC5 had a higher activity toward RHGR than RGKR, the short form of Lefty would be expected as the dominant form after exposure to PC5. Figure 1 shows that this is not the case.

(Choice D) If PC5 had equal activity toward both sequences, the long form of Lefty would not be the dominant form in the presence of PC5.

Educational objective:
Enzymes may act on multiple substrates and may show higher activity toward some substrates than others. These differences in activity can be measured empirically.

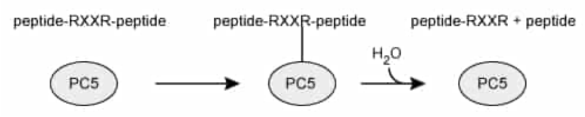
Time spent: 0 sec
Subject: Biochemistry

QID: 403361
System: Enzymes

Proprotein convertases (PCs) are serine proteases that cleave inactive proproteins at the surface of cells, converting them into active proteins. The proprotein convertase known as PC5 is critical for development, and knocking out PC5 during embryonic development is lethal in mice. After PC5 arrives at the cell surface, it first processes itself by autocleavage (ie, cleaving itself) before acting on any other proteins.

Many PCs cleave proteins at sites with the amino acid sequence RXXR, where X is any amino acid. In PC5, the side chain of a serine residue in the active site becomes deprotonated and acts as a nucleophile to form a covalent bond with an arginine residue, eventually hydrolyzing the protein at the C-terminal side of the bound arginine residue, as shown in Scheme A.

Scheme A



To characterize PC5 activity, a growth factor protein called Lefty was expressed in Chinese Hamster Ovary (CHO) cells. The inactive form of Lefty (Prolefty) has two PC5 cleavage sites, as illustrated in Figure 1. Depending on which site or sites PC5 cleaves, Prolefty may be converted into one of two Lefty isoforms: the long form (L) or the short form (S). Relative amounts of each form of Lefty, either in the absence or presence of PC5 coexpression, were measured in CHO cells.

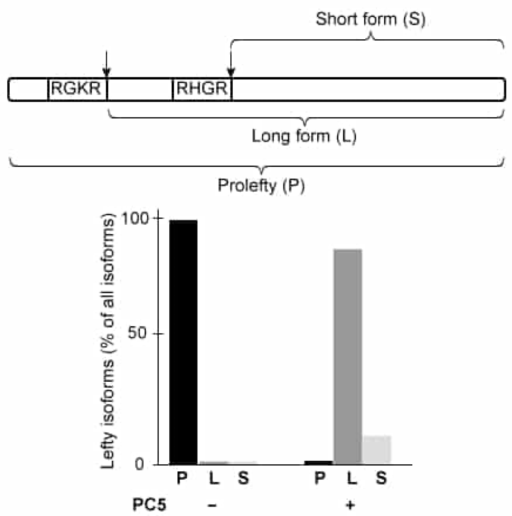


Figure 1 Schematic of Lefty (top) and graphical results of relative levels of each Lefty isoform detected by Western blot in the absence (−) or presence (+) of PC5 (bottom). Arrows in the top panel indicate cleavage sites.

Some chloromethylketone (CMK) derivatives are known to inhibit another PC called furin.

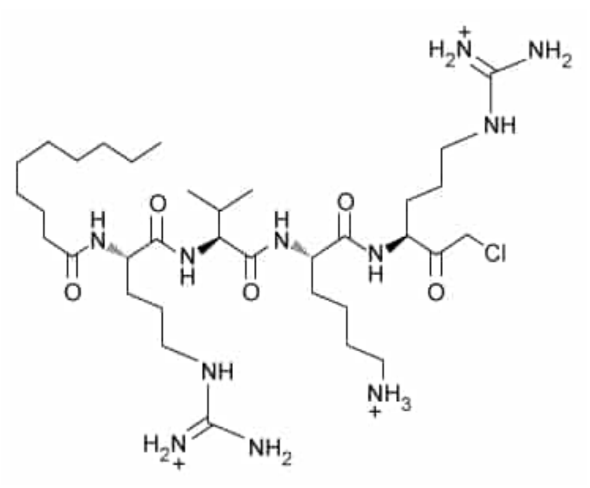


Figure 2 Structure of a CMK molecule used to inhibit furin.

In a separate experiment, PC5 activity against Lefty was assessed in a mouse cell line, either in the absence or presence of a specific CMK that inhibits furin (Figure 3).

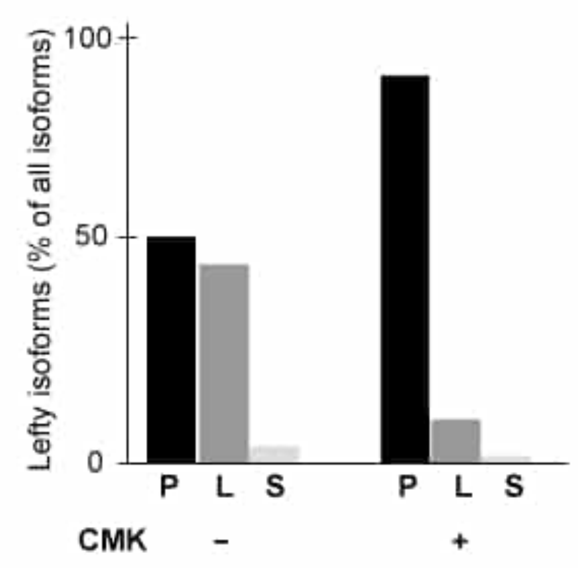


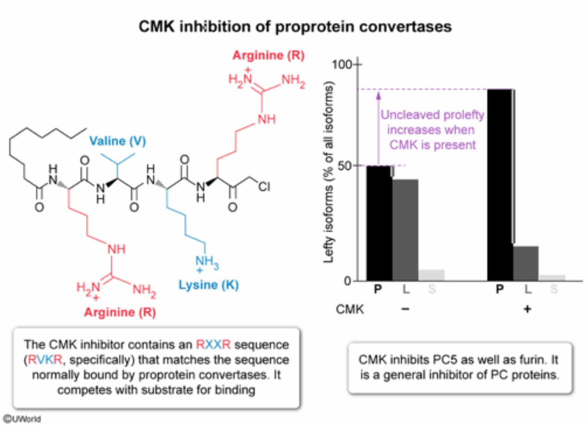
Figure 3 Graphical results of the relative amounts of each Lefty isoform in mouse cells expressing PC5 in the absence (−) or presence (+) of the furin-inhibiting CMK.

Which of the following best describes the CMK inhibitor used in Figures 2 and 3?

- A. It is a competitive inhibitor that is specific to furin. [32%]
- B. It is an allosteric inhibitor that is specific to PC5. [13%]
- ✔ C. It is a general competitive inhibitor of proprotein convertases. [34%]
- D. It is a general allosteric inhibitor of proprotein convertases. [20%]

Correct
34% answered correctly

Explanation:



Enzymes are often part of larger groups that all catalyze **similar reactions** by **similar mechanisms**. Consequently, these enzymes often have **similar active sites** that can be bound by the **same inhibitor**. Inhibitors that bind the active site of an enzyme are typically structurally similar to the substrate for that enzyme and **compete with the substrate** for binding. For this reason, such inhibitors are called competitive inhibitors.

The passage states that many PCs cleave proteins after the sequence RXXR, where X can be any amino acid. Figure 2 shows that the CMK inhibitor contains the amino acids RVKR, which fits the RXXR criteria for the active site. Therefore, CMK most likely inhibits furin competitively. Figure 3 shows that the inhibitor is also able to reduce the cleavage of Prolefty by PC5. Therefore, the CMK inhibitor is effective against both furin *and* PC5, and it is most likely a *general* competitive inhibitor of PC proteins rather than being specific to furin or to PC5 alone.

(Choice A) If the inhibitor were specific to furin, then Figure 3 would show no inhibition of PC5 in the presence of CMK. Instead, Figure 3 shows that CMK *can* inhibit PC5.

(Choices B and D) Allosteric inhibitors bind enzymes at a site other than the active site (ie, an allosteric site). Figure 2 shows that the inhibitor has a structure that includes the amino acid sequence that binds to PC active sites for catalysis. Therefore, it most likely binds to the active site rather than an allosteric site.

Educational objective:

Enzyme inhibitors often affect multiple similar enzymes. Competitive inhibitors have structures that can interact with the enzyme active site, allowing them to compete with substrate for binding interactions.

Time spent:0 sec **QID:**403362
Subject: Biochemistry **System:** Enzymes